

PAAMIUT FREDERIKSHAAB, Greenland

WMO: 042600

Lat: 62.015N

Lon: 49.671W

Elev: 15

StdP: 101.14

Time Zone: -3.00 (GRL)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-18.4	-16.2	-26.8	0.3	-15.9	-24.5	0.4	-13.7	17.9	-4.4	16.1	-4.1	3.0	20	0.867

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.4	13.6	8.6	11.9	8.0	10.6	7.4	9.6	11.8	8.8	10.8	8.0	9.9	5.2	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.4	6.9	10.2	7.5	6.4	9.3	6.7	6.1	8.6	28.6	11.9	26.6	10.9	24.9	10.1	16.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.7	12.7	11.0	-19.6	17.3	4.0	2.7	-22.5	19.2	-24.9	20.8	-27.1	22.3	-30.0	24.2
			-20.6	11.5	3.7	1.6	-23.2	12.6	-25.3	13.5	-27.4	14.4	-30.0	15.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.1	-6.0	-6.7	-5.8	-1.3	2.2	4.9	6.6	6.5	4.5	1.1	-2.5	-4.7
	DBStd	6.16	5.00	5.53	5.32	3.82	2.86	2.31	2.24	2.24	2.25	3.20	4.20	4.72
	HDD10.0	3679	495	467	489	338	243	154	110	111	166	275	376	457
	HDD18.3	6712	753	701	747	588	500	403	365	367	415	533	626	715
	CDD10.0	8	0	0	0	0	1	1	3	2	1	1	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	4.3	4.7	5.1	4.9	4.8	4.3	3.6	3.3	3.5	4.4	4.2	4.4	4.5
Precipitation	PrecAvg	1333	138	134	107	91	79	76	75	100	133	114	151	135
	PrecMax	2238	366	345	289	235	198	150	183	224	262	263	327	350
	PrecMin	742	37	35	15	18	25	5	7	28	38	41	33	34
	PrecStd	290	78	81	60	47	36	40	43	45	54	59	81	72
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.7	9.1	9.0	10.3	14.2	13.8	15.7	15.1	14.4	13.1	10.7	8.4
		MCWB	3.8	5.7	4.5	4.0	7.8	8.1	10.6	10.4	8.0	7.1	5.7	3.2
	2%	DB	5.6	6.0	6.2	7.5	10.1	11.4	12.9	12.5	10.9	9.0	7.0	5.5
		MCWB	1.6	2.0	2.6	3.2	6.2	7.2	8.9	9.0	6.9	4.6	3.3	1.9
	5%	DB	2.8	3.3	3.7	5.6	7.8	9.7	11.4	11.0	9.0	6.9	4.4	3.5
		MCWB	0.0	0.1	0.8	2.7	4.8	6.8	8.2	8.4	6.6	3.9	1.7	1.0
	10%	DB	1.0	1.1	1.7	3.8	6.2	8.4	10.0	9.8	7.7	5.4	2.7	1.8
		MCWB	-1.2	-1.0	-0.3	1.6	4.0	6.1	7.7	7.8	5.6	3.2	0.8	-0.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.6	5.6	5.6	5.7	8.8	9.2	11.2	11.0	9.3	7.5	6.4	4.6
		MCDB	7.7	8.9	8.0	7.9	12.1	12.1	14.2	14.1	11.8	11.1	10.1	6.7
	2%	WB	2.0	2.7	3.0	4.2	6.6	8.0	9.6	9.6	7.9	5.5	4.1	2.6
		MCDB	4.8	4.7	5.5	6.3	9.7	10.2	11.7	11.5	10.1	7.5	6.8	4.7
	5%	WB	0.5	0.8	1.1	2.7	5.2	7.1	8.6	8.9	6.8	4.3	2.2	1.2
		MCDB	2.6	2.9	3.3	4.7	7.4	9.2	10.6	10.6	8.6	6.1	3.5	3.2
	10%	WB	-1.1	-0.9	-0.4	1.7	4.2	6.3	7.8	8.0	5.9	3.4	1.0	0.0
		MCDB	0.7	0.8	1.5	3.6	5.8	8.2	9.7	9.5	7.4	5.1	2.5	1.6
Mean Daily Temperature Range	5% DB	MDBR	5.3	5.7	5.7	4.9	4.0	4.4	4.4	4.0	3.9	4.5	4.9	5.0
		MCDBR	7.8	8.1	8.6	7.9	7.4	7.2	7.3	6.4	6.7	6.8	7.2	6.6
		MCWBR	5.5	5.6	6.0	5.0	4.2	4.1	4.2	3.7	3.9	4.2	5.2	4.4
	5% WB	MCDBR	7.7	8.0	8.4	7.0	6.5	6.5	6.5	5.8	5.9	6.0	6.6	6.2
		MCWBR	5.8	6.2	6.2	5.0	4.2	4.1	4.2	3.8	3.9	4.3	5.3	4.6
Clear-Sky Solar Irradiance	taub	0.235	0.272	0.292	0.334	0.349	0.352	0.360	0.357	0.322	0.284	0.227	0.366	
	taud	2.027	2.249	2.326	2.261	2.325	2.391	2.385	2.418	2.451	2.365	2.092	2.794	
	Ebn at Noon	438	702	827	852	868	872	854	825	790	682	452	323	
	Edh at Noon	37	61	82	108	110	106	104	92	74	54	33	22	
All-Sky Solar Radiation	RadAvg	0.23	0.91	2.41	4.30	5.66	6.16	5.70	4.27	2.64	1.37	0.39	0.12	
	RadStd	0.02	0.09	0.20	0.34	0.28	0.37	0.45	0.33	0.25	0.10	0.05	0.01	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	+0.61	+2.90	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (2 neighbors)	+0.61	+2.90	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page